



**RAJALAKSHMI ENGINEERING COLLEGE,
CHENNAI**

**Department of Artificial Intelligence & Data Science
CS23633 – Cloud Computing Laboratory**

Experiment 3

**Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer
(Simulation)**

AIM:

To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer

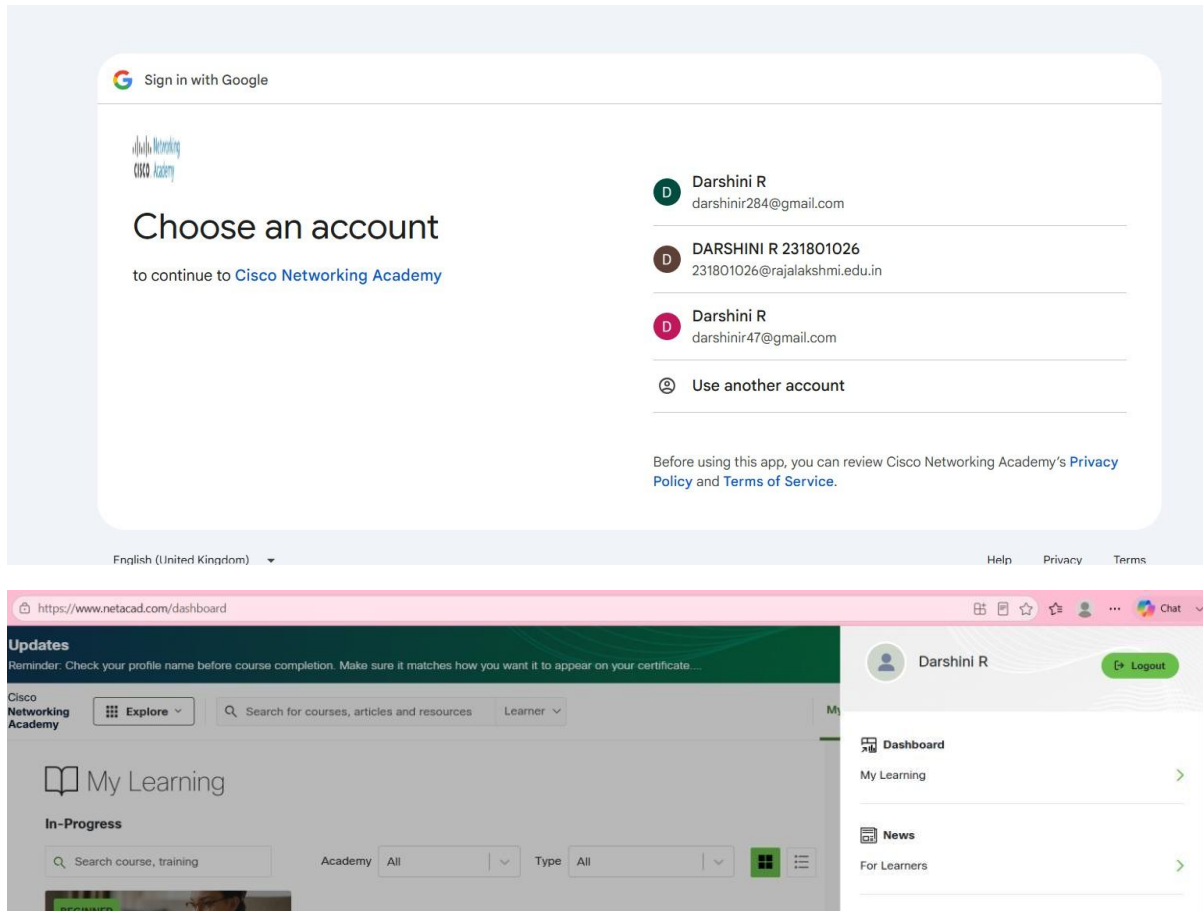
PROCEDURE:

Step 1: Open Cisco Networking Academy

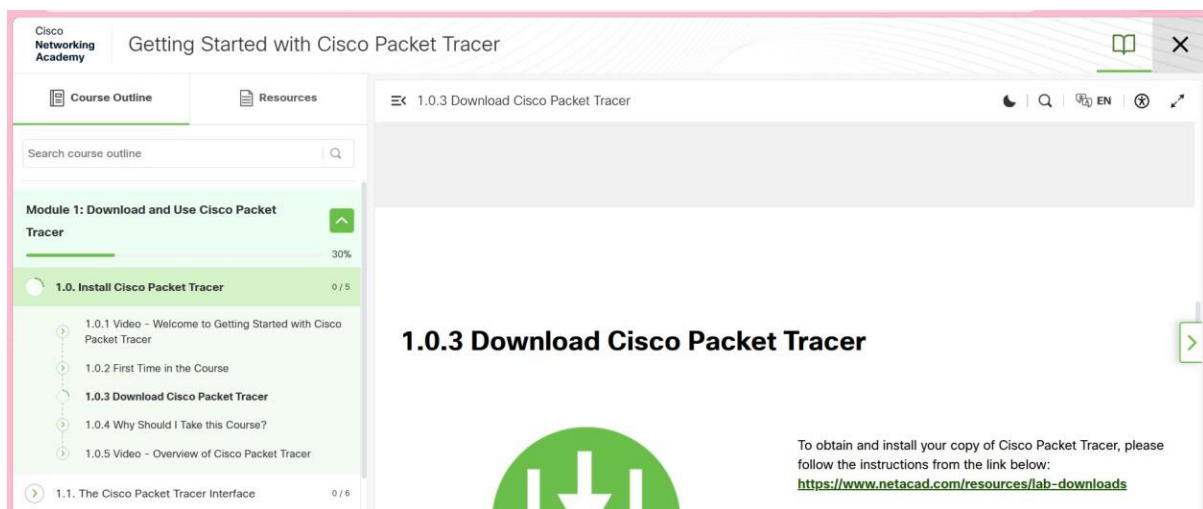
Go to: <https://www.netacad.com>

Step 2: Create a Free Account or Sign in with your Google Account

1. Click Create an account (or Sign up).
2. Enter:
 - o First Name
 - o Last Name
 - o Email Address
 - o Password
3. Verify your email if prompted.
4. Sign in.



Step 3: Enroll in the Free Packet Tracer Course

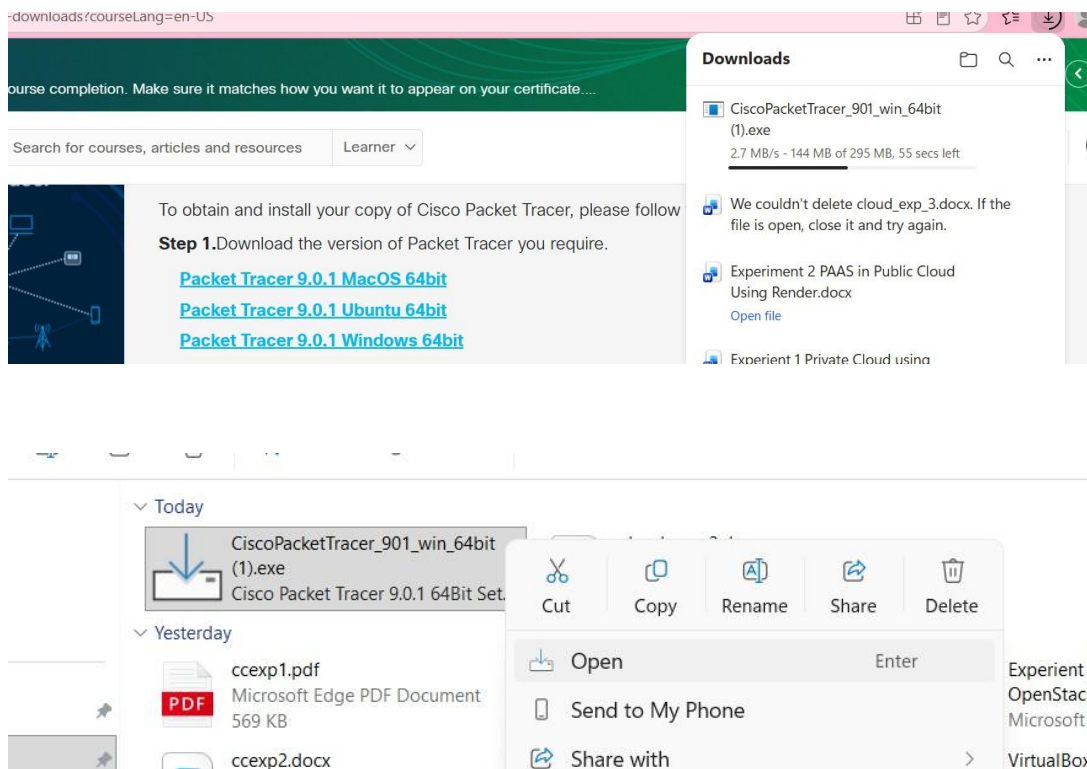


Step 4: Download Packet Tracer

After enrolling:

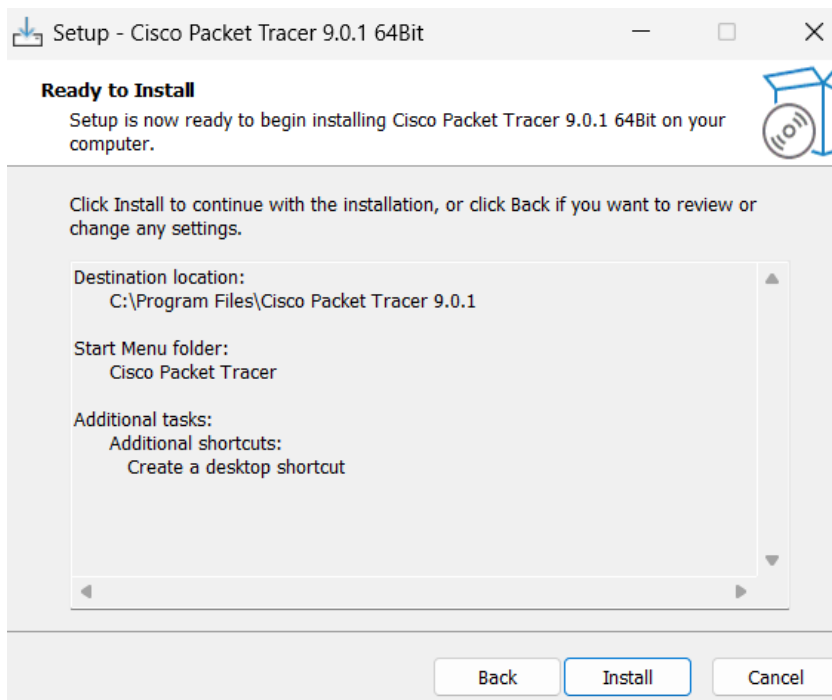
1. Go to your NetAcad Dashboard.

2. Open the Getting Started with Cisco Packet Tracer course.
3. Navigate to the Resources or Downloads section.
4. Download the version for your operating system:
 - o Windows 64-bit (recommended for most PCs)
 - o macOS
 - o Linux



Step 5: Install Packet Tracer

1. Open the downloaded installer (.exe on Windows).
2. Click **Next**.
3. Accept the license agreement.
4. Click **Next** until installation completes.
5. Click **Finish**.



Step 6: Start Packet Tracer

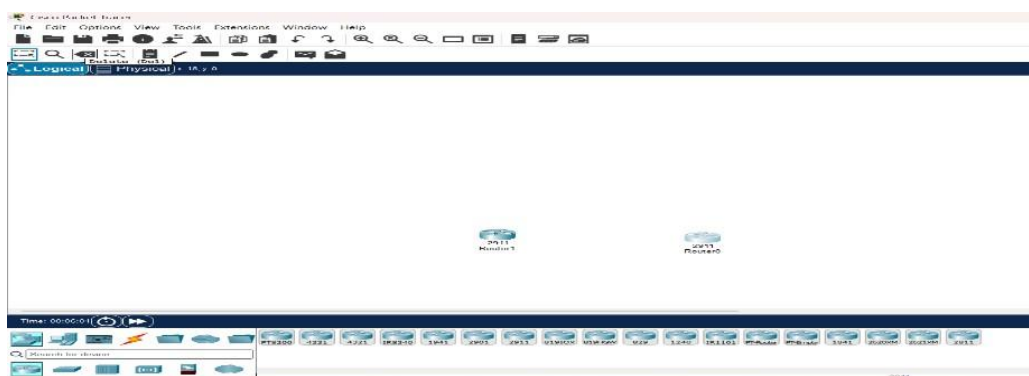
Open **Cisco Packet Tracer** from the Start Menu.

Sign in with your **Cisco NetAcad** account or sign in with your google account.

Step 7: Place Two Routers

At the **bottom-left** of Packet Tracer, you'll see several icons.

1. Click **Network Devices** (the router icon).
2. Click **Routers**.
3. Choose **2911** (or **1941** if 2911 isn't available).
4. Click once in the blank workspace to place the first router and second one



Step 10: Connect the Devices

At the bottom-left, click the **Connections** icon  (lightning bolt). Choose

Copper Straight-Through (usually the first black cable). **Connection 1:**

PC to Switch0

1. Click **PC**.
2. Select **FastEthernet0**.
3. Click **Switch0**.
4. Select **FastEthernet0/1** (or any free FastEthernet port).

Connection 2: Switch0 to Router0

Using the same **Copper Straight-Through** cable:

1. Click **Switch0**.
2. Select **FastEthernet0/2**.
3. Click **Router0**.
4. Select **GigabitEthernet0/0**.

Connection 3: Server to Switch1

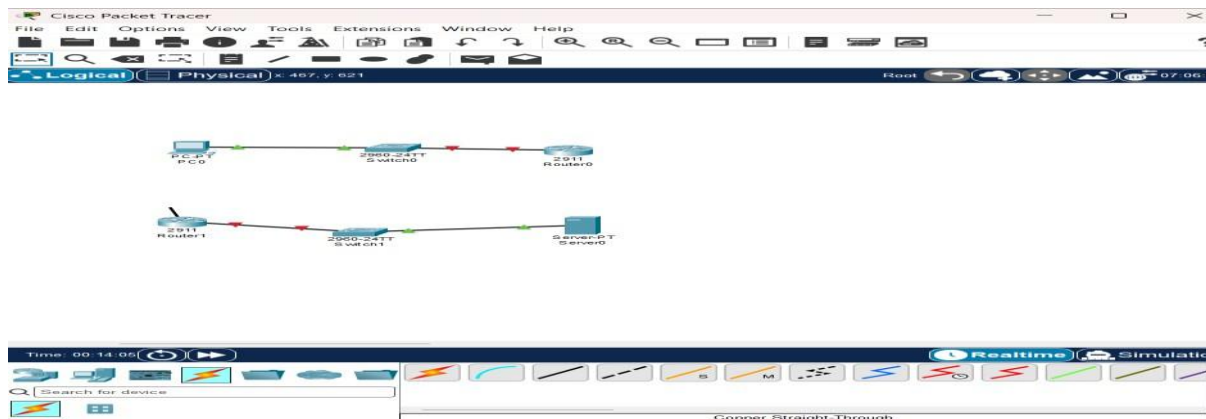
Again use **Copper Straight-Through**:

1. Click **Server**.
2. Select **FastEthernet0**.
3. Click **Switch1**.
4. Select **FastEthernet0/1**.

Connection 4: Switch1 to Router1

Again use **Copper Straight-Through**:

1. Click **Switch1**.
2. Select **FastEthernet0/2**.
3. Click **Router1**.
4. Select **GigabitEthernet0/0**.



Step 10.1: Connect Router0 to Router1

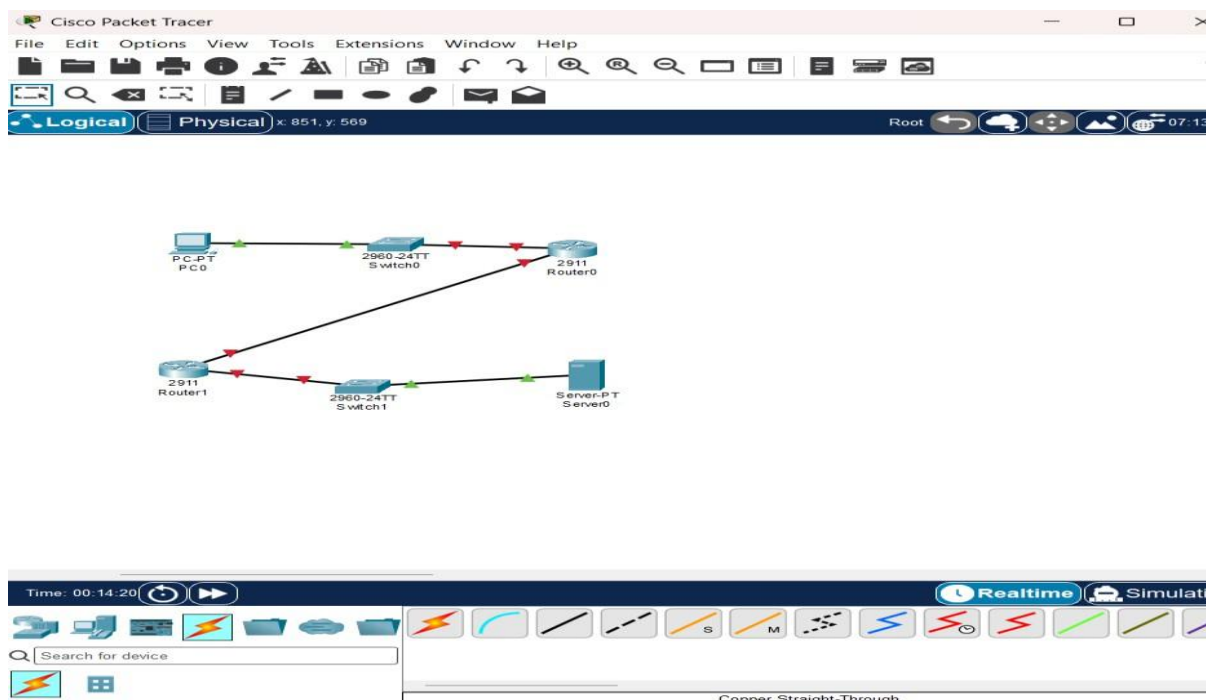
Now connect the two routers.

Option A (Recommended)

Use Copper Cross-Over.

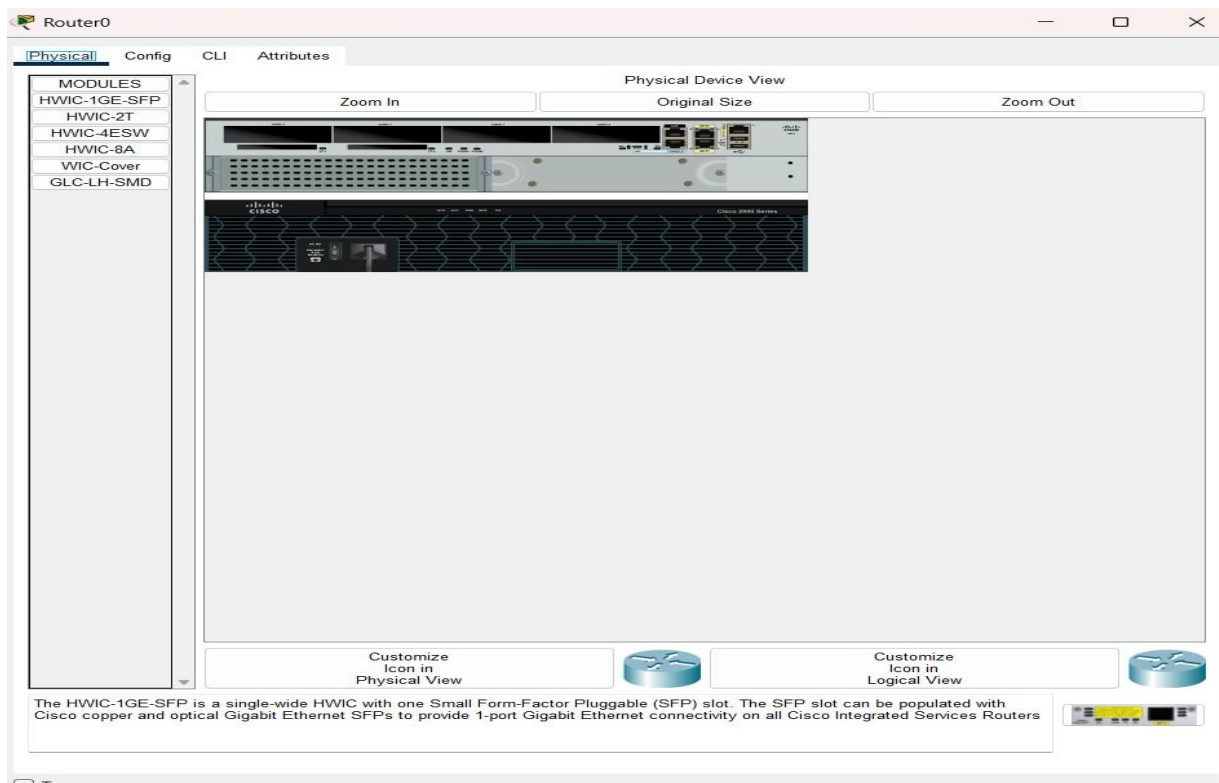
1. Click the **Connections** icon.
2. Select **Copper Cross-Over**.
3. Click Router0.
4. Select **GigabitEthernet0/1**.
5. Click **Router1**.
6. Select **GigabitEthernet0/1**.

Your topology should now look like this:



Step 10.2: Click the Router

In your workspace, **left-click Router0** once.



1. Click the CLI tab.

2. If it asks:

Continue with configuration dialog? [yes/no]:

Type: no Press **Enter**. When you see: Router>

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

smart init is sizing iomem
      TYPE      MEMORY_REQ
Onboard devices &
  buffer pools 0x022F6000
-----
TOTAL:         0x022F6000
Rounded IOMEM up to: 36Mb.
Using 6 percent iomem. [36Mb/512Mb]

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170 West Tasman Drive
San Jose, California 95134-1706

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http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

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Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
  
```

Step 10.3: Configure the LAN Interface (Connected to Switch0)

At the Router(config)# prompt, type these commands **one at a time**, pressing **Enter** after each line.

```
interface    gigabitEthernet0/0
```

You should see:

```
Router(config-if)#
```

Now type:

```
ip address 192.168.1.1 255.255.255.0
```

Press **Enter**.

Now type:

```
no shutdown
```

You should see a message similar to: Now

```

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>configure terminal
      ^
% Invalid input detected at '^' marker.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip address 192.168.1.1 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#no shutdown
      ^
% Invalid input detected at '^' marker.

Router(config)#Router0
      ^
% Invalid input detected at '^' marker.

Router(config)#prompt
      ^
% Invalid input detected at '^' marker.

Router(config)##prompt
      ^
% Invalid input detected at '^' marker.

Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

Step 10.4: Configure the Router-to-Router Interface

Now type:

```
interface    gigabitEthernet0/1
```

Then:

```
ip address 10.10.10.1 255.255.255.252
```

Then:

```
no shutdown
```

Again, wait for the interface to come up.

Finally, type:

```
exit
```

You should now be back at:

```
Router(config)#
```

```

Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

```

Step 10.5: Verify the Configuration

Type:

do show ip interface brief

Since you're in configuration mode, use **do** before the show command.

You should see something similar to:

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up             up
GigabitEthernet0/1       10.10.10.1     YES manual up             up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router(config)#ip route 10.0.0.0 255.255.255.0 10.10.10.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#

```

Next Step: Configure Router1

Click **Router1** → **CLI**.

If it says:

Press RETURN to get started!

Press **Enter**.

If it asks:

Continue with configuration dialog? [yes/no]:

Type:

no

Then enter these commands **one by one**: enable

configure terminal

interface gigabitEthernet0/0

ip address 10.0.0.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet0/1

ip address

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

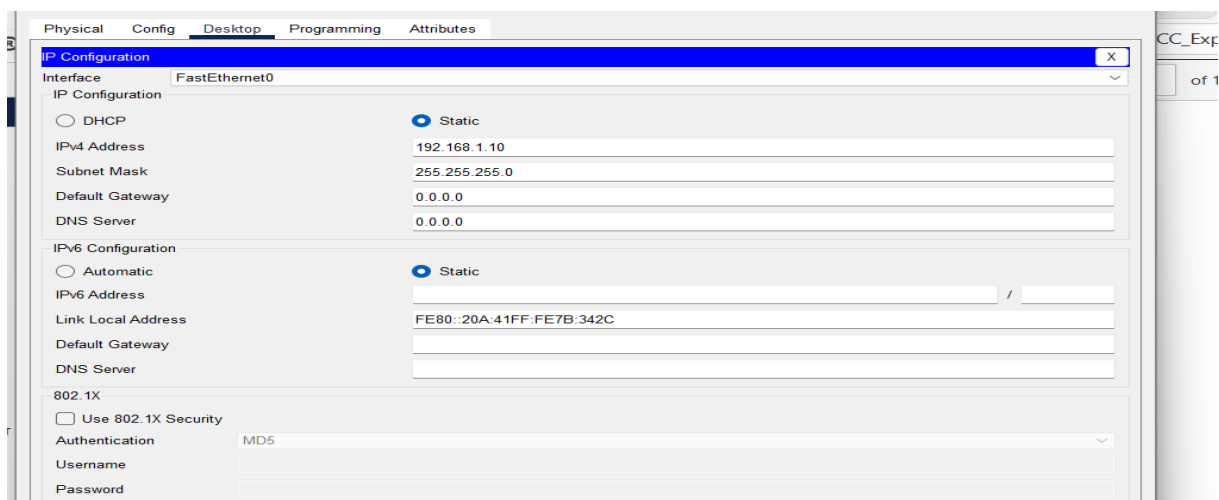
Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
Router(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    10.0.0.1        YES manual  up          up
GigabitEthernet0/1    10.10.10.2      YES manual  up          up
GigabitEthernet0/2    unassigned      YES unset   administratively down down
Vlan1            unassigned      YES unset   administratively down down
Router(config)#
```

Step 10.6: Configure the PC

1. Click PC.
2. Select **Desktop**.
3. Click **IP Configuration**.



Step 10.7: Configure the Server

1. Click Server.

2. Select **Desktop**.

3. Click **IP Configuration**.

The screenshot shows the 'Server0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 10.0.0.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.0.0.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: [empty field] / [empty field]
 - Link Local Address: FE80::290:CFF:FEDD:9749
 - Default Gateway: [empty field]
 - DNS Server: [empty field]
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MD5
 - Username: [empty field]
 - Password: [empty field]

At the bottom left, there is a 'Top' button.

Step 11: Configure Static Routes

A. Configure Static Route on Router0

1. Click **Router0**.

2. Go to the CLI tab.

3. If you're at Router#, type:

configure terminal

Now enter:

```
ip route 10.0.0.0 255.255.255.0 10.10.10.2
```

Exit configuration mode: exit

Save the configuration: copy running-config startup-config

When prompted: Destination filename [startup-config]? Just press **Enter**.

B. Configure Static Route on Router1

1. Click **Router1**.

2. Open the CLI.

At the Router# prompt, type: configure terminal

Now enter: ip route 192.168.1.0 255.255.255.0 10.10.10.1

Exit: exit

Save the configuration: copy running-config startup-config When prompted for the filename, simply press **Enter**.

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up          up
GigabitEthernet0/1       10.10.10.1      YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router(config)#ip route 10.0.0.0 255.255.255.0 10.10.10.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

C. Verify the Route

Type: show ip route

You should see a line similar to: S 192.168.1.0/24 [1/0] via 10.10.10.1 The

S indicates that the static route has been added successfully.

```
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.0.0/24 [1/0] via 10.10.10.2
C       10.10.10.0/30 is directly connected, GigabitEthernet0/1
L       10.10.10.1/32 is directly connected, GigabitEthernet0/1
C       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
Router#
```

Copy

Paste

Now let's verify Router0

On **Router0**, at the Router# prompt, type: show ip route You should see a line like: S 10.0.0.0/24 [1/0] via 10.10.10.2

If you don't see it, add the route:

configure terminal ip route 10.0.0.0 255.255.255.0 10.10.10.2

end copy running-config startup-config

Press **Enter** when it asks for the destination filename.

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

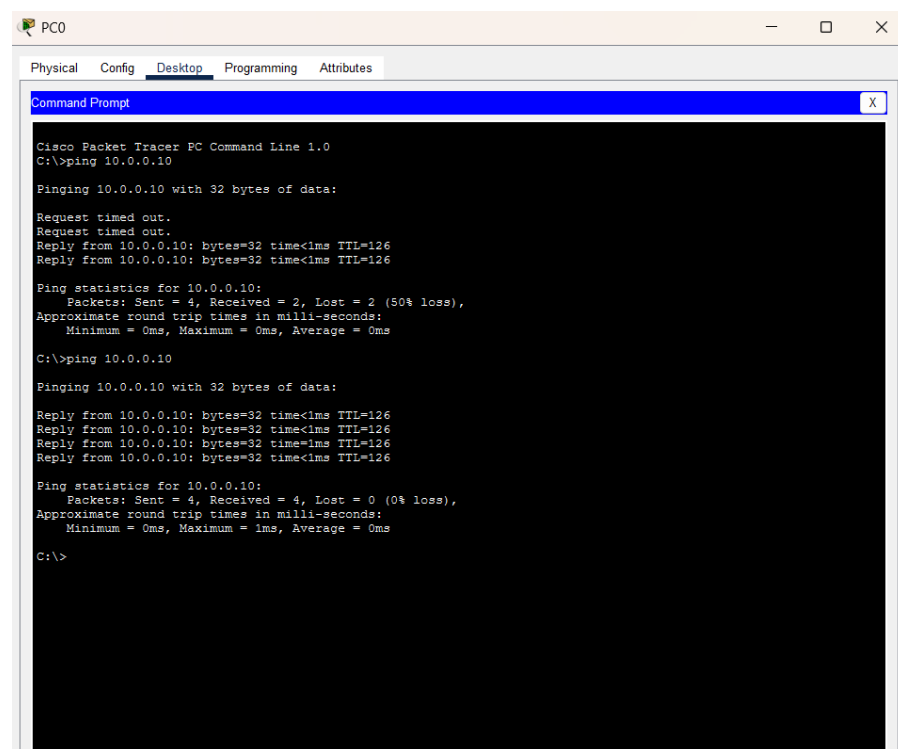
    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.10.10.0/30 is directly connected, GigabitEthernet0/1
L       10.10.10.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 10.10.10.1

Router#
```

Step 12: Test Connectivity

From the PC

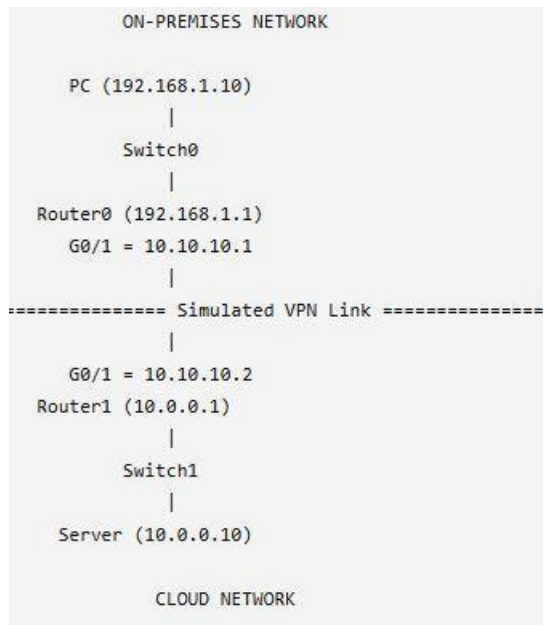
1. Click PC.
2. Go to **Desktop** → **Command Prompt**.
3. Type: ping 10.0.0.10
4. Type one more time: ping 10.0.0.10



Experiment Successfully Completed!

You have successfully simulated **Hybrid Cloud Connectivity**.

Architecture Diagram of Simulation



Result:

A hybrid cloud network was successfully simulated using Cisco Packet Tracer. The on-premises network and cloud network were interconnected through routers configured with static routing. Communication between the PC and the cloud server was verified successfully using the ping command, demonstrating end-to-end connectivity.

